

VOLLEY | Final B.Tech Project Review

Panel handbook: completed Gen5 study, findings, and future direction

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Scope of the completed thesis. VOLLEY is a system-level engineering design and computational research project. The Gen5 analysis, comparisons and decision record are complete for the stated models. The results are predictions, with numerical cross-checks where identified. Physical prototype testing, CubeSat qualification and provider accommodation belong to a later development phase. Gen6 is presented separately as future work.

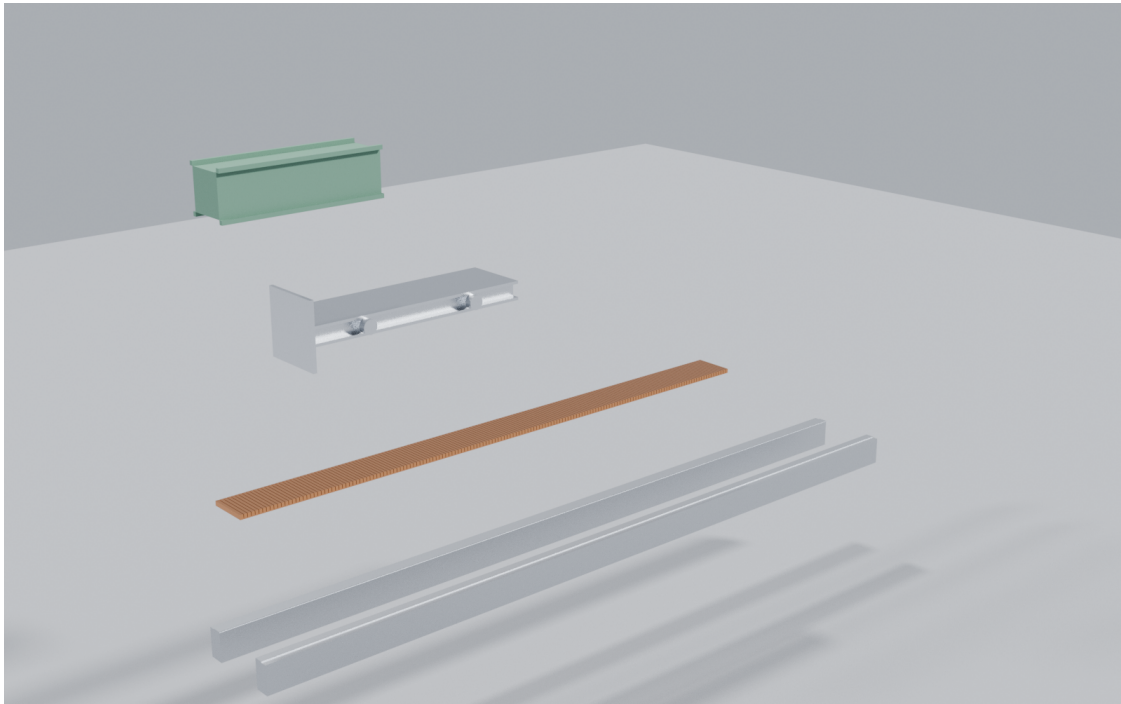


Figure 1: Analysed Gen5 electromagnetic deployer configuration. CAD render; no physical article exists.

Executive summary

VOLLEY investigates a last-mile problem in rideshare launches: a secondary CubeSat reaches orbit but ordinarily receives the orbit selected for the primary payload. A spring deployer separates it safely, often at roughly 1–2 m/s; that relative impulse, later drag differences, or host maneuvering can change its in-track phase. Deployment time by itself does not create persistent phase separation from an otherwise unchanged co-orbital host. A larger, deliberately commanded release velocity can also change orbital energy. The thesis evaluates whether that added capability justifies a complete deployer system. The mission model splits responsibilities between a maneuvering host, which performs coarse repositioning, and a deployer, which supplies each satellite's commanded release condition.

The authored thesis analyses one Gen5 solution: a magazine-fed, linear synchronous motor driving a reusable magnetic sled. The rated 3U model predicts **16.0 m/s** exit speed at **10.07 g**. Its full dry system mass is **126.6 kg**, and the power model predicts **2.782 kJ** gross electrical draw per shot. A closed-loop simulation predicts **0.0274 m/s** (3σ) exit-speed dispersion at a 15.8 m/s setpoint. In a stated orbital case, a maximum-speed release raises semi-major axis by **28.8 km**, with a **1.60×** modelled lifetime multiplier at mean solar activity. The lifetime result is atmosphere-sensitive and has not been independently rerun at the exact current operating point.

Gen5 also produced a decisive negative result. With twelve 3U satellites, its deployer mass is **10.55 kg per customer**, versus about **6 kg** for the spring canister comparator used in the thesis. This is **1.76×** as much hardware per 3U payload and fails the preset 15% mass-parity band. The study therefore reaches a decision: the modelled orbital benefit does not make this Gen5 configuration a defensible 3U selection. The result completes the analytical comparison; it does not require a prototype to know that the modelled mass criterion failed. Physical payload loads, magnetic cleanliness, cycle life, and provider interfaces remain questions for a different evidence phase.

The later Gen6 programme retains the hosted sequential-deployment mission while reopening mechanism choice. It investigates an approximately **1–2 m/s to 1 km/s** release-speed envelope across electromagnetic, mechanical, stored-energy, gas/fluid, hybrid, BOLLEY and conventional options. The high end is a requirement to examine and may prove infeasible. The near-term goal is one named, reproducible, independently reviewable test-article configuration with a cor-

responding build/test release decision.

1. Introduction

The deployment problem

The primary mission determines the launch vehicle's delivered orbit. Secondary CubeSats may have limited onboard propulsion, mass, power, or programmatic freedom to correct that orbit. Standard canisterised spring systems are mature and effective at clearance and separation. Their release conditions, however, are governed by spring energy, payload mass, and the device rather than a separately commanded velocity profile for every satellite.

The distinction between **release timing** and **relative-state change** is central. Waiting alone between zero-relative-impulse releases from an unchanged host does not produce persistent in-track phase spacing. Spring impulse, differential drag or a host maneuver can produce a relative state; a commanded velocity along or against orbital motion can change the satellite's orbital energy and semi-major axis. VOLLEY studies the latter capability while retaining a clear comparison with spring release and other alternatives. It does not claim that all constellation phasing requires a motor.

The host is an active participant in the mission model. After its primary mission, a suitable hosted platform would need attitude control, available resources and permission to operate. It could perform coarse orbital repositioning; VOLLEY would assign each secondary payload a local release state. No specific upper stage has been selected or approved for that role.

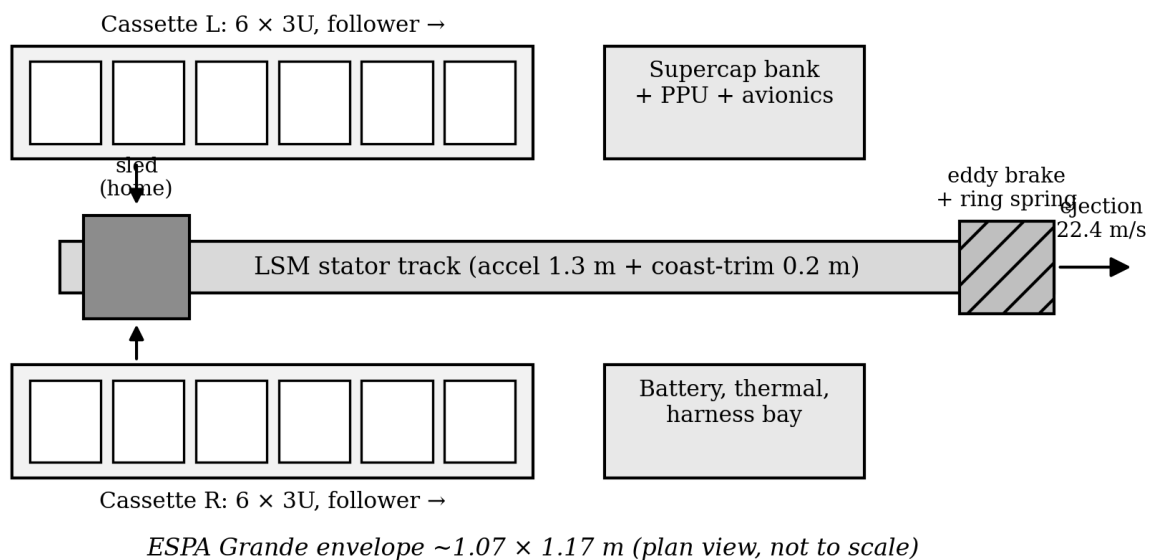


Figure 2: Gen5 layout and mission hardware elements. Diagram from the thesis analysis.

2. Literature review

Approach	Established capability	Relevance and limit
Spring deployers	Flight heritage and simple separation near the low-m/s range [1]	Strong benchmark for reliability and installed mass; release energy is largely built into the device.

Approach	Established capability	Relevance and limit
Spring release and differential drag	In-track spacing and gradual orbital adjustment [3]	A relative impulse or drag difference creates the separation; waiting alone from an unchanged co-orbital host does not.
Orbital transfer vehicles	Coarse orbital repositioning through a propulsive carrier	Valuable comparator for whole-mission delivery; greater spacecraft and integration burden must be counted.
Electromagnetic CubeSat studies	Magazine transport and electromagnetic launch have already been studied [4–6]	Some configurations impose conductive armatures, high acceleration or different maturity; novelty cannot be claimed for electromagnetic deployment alone.

The most relevant high-speed electromagnetic comparison in the manuscript is Feng *et al.* [4], which models a 321.56 m/s launch of a 20 kg CubeSat but at acceleration on the order of thousands of g and with an armature/interface issue. Zhao *et al.* [5,6] study stacked CubeSat transfer and storage, including measured transport behavior in a related mechanism. These works establish that electromagnetic delivery and magazine transport are prior art. VOLLEY's more limited question is whether a **commandable** release can be delivered to an **unmodified** satellite at a much lower acceleration while accounting for the whole installed system. No universal CubeSat quasi-static acceleration qualification is assumed; the launch provider and payload-specific limits must govern a real design [2,8].

3. Research gap

The analytical gap is not “no one has considered electromagnetic deployment.” It is the coupled trade between: (i) commanded release, (ii) payload-side interface and acceleration, (iii) a reusable feed/release architecture, (iv) host reaction and mission utility, and (v) installed system mass and faults. A speed-only comparison can make a concept appear attractive while omitting its sled, power bank, brake, magazine, enclosure, control, and retained mass after each shot.

The thesis addresses that gap by constructing a complete Gen5 computational configuration and testing its own claims against explicit comparisons. It also records when those tests weaken the concept. Physical viability remains a separate research gap: numerical model agreement cannot establish manufactured tolerances, actual force, payload cleanliness, cycle reliability or provider compatibility.

4. Objectives and outcome

Analytical research objective	Completed outcome	Interpretation
Model a commandable 3U system	Gen5 CAD, magazine, sled, motor, arrest and power architecture	System-level design record, not manufactured hardware
Quantify the release	Rated shot, circuit and closed-loop simulations	Predicted performance under stated assumptions
Evaluate orbital value	Orbit model and selected independent numerical cross-checks	Conditional energy and lifetime benefit

Analytical research objective	Completed outcome	Interpretation
Test whole-system competitiveness	Mass rollup, spring/alternatives comparison and preset criteria	Gen5 fails the 3U mass target
Challenge model credibility	Numerical verification, defect corrections and provenance record	Scope and remaining physical uncertainties made explicit

These objectives summarize the completed analytical research for this final-review presentation. They are a presentation synthesis, not a claim that the university approved this exact list in advance. The physical development tasks are shown later under future work.

5. Research methodology

The project follows an engineering evidence chain:

1. **Mission/use-case framing:** specify the payload, host responsibilities, target orbital change, safety limits and competent conventional comparators.
2. **Architecture trade:** examine force production, loading, restraint, guidance, arrest, reset and fault behavior, rather than only ideal acceleration.
3. **Geometry and installed mass:** build CAD geometry and count the sled, arrays, enclosure, feed, power, controls and other installed parts.
4. **Coupled models:** compute magnetic field and thrust, shot dynamics, power/circuit behavior, control dispersion, structural/flow responses and orbital consequences.
5. **Challenge the claims:** use alternate numerical methods and acceptance bands, preserve misses and update the baseline only with a recorded reason.
6. **Decision:** compare orbital benefit against mass, reliability, payload/host compatibility and the available alternatives.

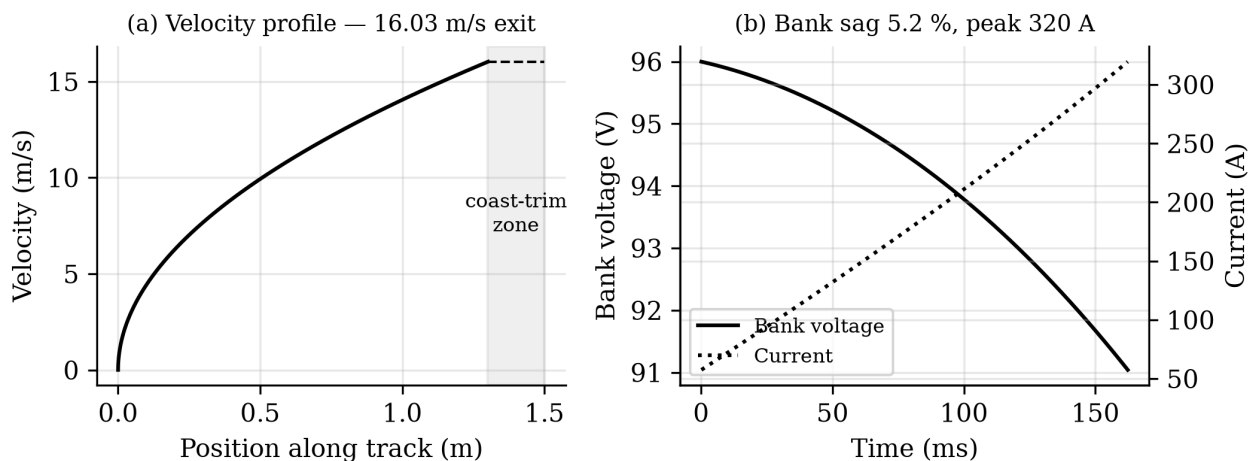


Figure 3: Modelled shot: velocity, bank voltage and current at the rated point.

Numerical checks and their limits

The magnetic field was compared across an analytical representation, magpylib, and 2-D/3-D finite-element calculations. A 3-D midgap solve supports the field model at its tested location but does not independently measure the depth-integrated thrust constant. Shot and circuit predictions were compared with ngspice at stated assumptions. Structural calculations use CalculiX; airflow calculations use OpenFOAM. Orbital calculations include Cowell and GMAT cases. The independent orbit work rejected an earlier assertion that lifetime benefit was invariant across solar activity. Each check applies to its declared case and boundary conditions; none is a hardware test.

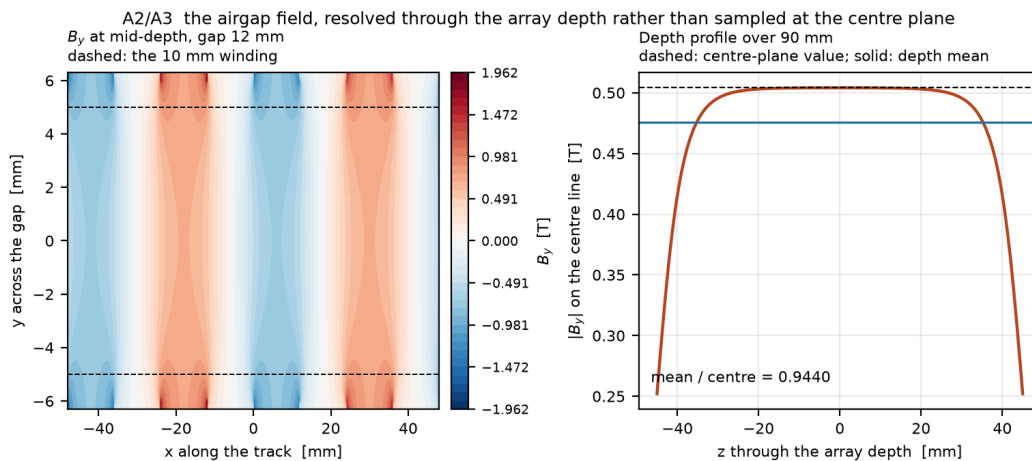


Figure 4: Representative magnetic-field model output. Source: thesis figure A02.

6. Timeline

Documented evolution

Period	Milestone	Confidence in date
2023	Free-flyer idea reframed around a host platform	Documented design decision
Mid-2025	Coilgun concept shifted toward a linear synchronous motor	Approximate historical milestone
2025–July 2026	CAD generations and mass model developed	Some dates reconstructed/approximate
July–September 2026	Gen5 analysis, numerical checks, defects and thesis record	Dated repository/analysis records
October 2026	Gen5 findings and design decision presented for final review	Review package dated 10 October 2026

The thesis repository’s reconstructed history expressly marks several early milestone dates as approximate. The analytical baseline was revised as evidence improved: a CAD-derived sled mass displaced a lighter parametric estimate; a depth-resolved thrust calculation lowered the rated speed; an enclosure buildup exposed the 3U mass penalty; and an independent orbit check rejected an overbroad solar-invariance claim. These revisions are part of the completed study, not signs that the final numbers are placeholders. Gen6 research gates are described only in §10, Future work.

7. Work progress

The thesis deliverables are in hand: an authored manuscript, CAD renders and model geometry, analysis scripts/results, numerical verification run sheets, figures, a baseline record, a defect register, and a provenance statement. Numerical investigations cover the motor field, shot dynamics, bank/circuit, mass, orbit propagation, separation and selected structural, airflow and reliability issues. The study closes by retaining failed criteria and superseded results rather than silently replacing them.

The deliverable here is a completed computational thesis, not a flight article. The later long gas-guide and independent spring-cell studies are historical comparators. Gen6 has a mission direction and investigation envelope but is not a result of the Gen5 thesis. Its mechanism and exact payload/provider cases remain future choices.

8. Results

Gen5 configuration

The analysed system uses a double-sided Halbach-array linear synchronous motor to propel a reusable permanent-magnet sled along a 1.5 m track. Two transverse cassettes feed twelve 3U satellites. A capacitor-based bank supplies the pulse, and a brake arrests the sled after payload release. The sled mass, computed from detailed CAD solid volumes, is **9.45 kg**; this value reduced the exit speed relative to an earlier optimistic parametric estimate. The CAD render conveys the arrangement; it is not evidence that parts have been manufactured or assembled.

Rated 3U shot

Model output	Rated result	Interpretation
Exit velocity	16.029 m/s	Modelled 3U reference shot
Payload acceleration	10.07 g	Not a payload qualification claim
Pulse duration	162.3 ms	Modelled motor actuation
Peak current	320 A	Component implementation unverified
Electrical energy drawn, gross	2,782 J	Bank/circuit model
Recovered after release	47 J	39 mm available recovery zone
Sled energy to brake	1,162 J	Major loss and arrest burden
Electrical-to-payload efficiency, net	18.8%	514 J payload kinetic energy / net draw

The energy flow matters more than the velocity headline. The sled and its arrest hardware are the price of keeping the customer satellite unmodified. Regeneration returns only a small fraction of sled energy within the available length, so the contactless brake remains necessary. A design comparison that credits full sled recovery would misstate the system efficiency and installed burden.

Command precision

The closed-loop Monte Carlo model predicts **0.0274 m/s (3σ)** exit-velocity dispersion around a **15.8 m/s** fleet setpoint. It uses assumed sensing and plant tolerances, with headroom below the open-loop ceiling. This is a simulated distribution, not a measured release repeatability.

Orbital utility

For the stated orbit and mean-solar-activity case, one maximum-speed release predicts a **28.8 km** semi-major-axis increase and a **1.60×** lifetime multiplier for a propulsionless satellite. The more defensible message is conditional: a release impulse can change orbital energy, while absolute lifetime and lifetime gain depend on atmospheric density and the orbit model. A fresh independent run of the exact current 1.60 operating point remains outstanding. Timing alone, with zero relative impulse and no host or drag difference, produces neither persistent phasing nor this semi-major-axis change.

Installed mass changes the decision

The dry Gen5 rollup is **126.6 kg** and loaded mass **174.6 kg**. At twelve 3U customers, the deployer weighs **10.55 kg per customer**, compared with approximately **6 kg** for the spring canister comparator used in the analysis. The ratio is **1.758**; the preset acceptance band required parity within 15%. That band fails by a large margin. Replacing an **8 kg enclosure placeholder** with a **50.04 kg** geometry-derived buildup exposed much of the penalty. Even changing structure alone does not recover parity in the studied options. The result is a reason to reopen the architecture, not to conceal the comparison.

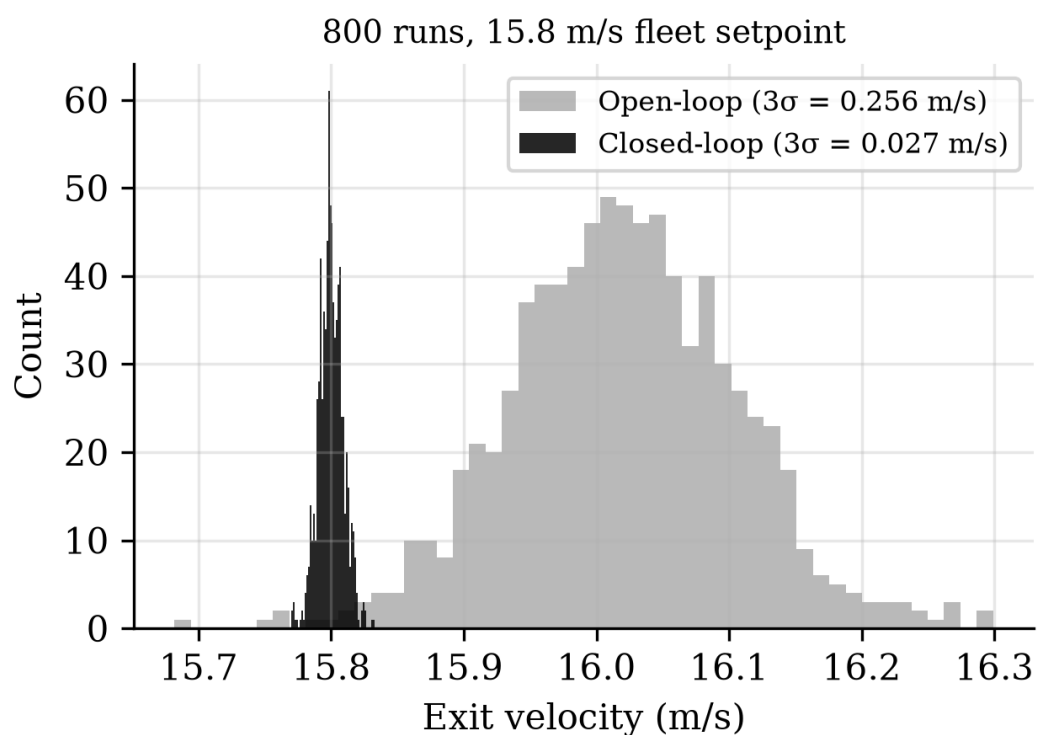


Figure 5: Closed-loop simulation distribution at the fleet setpoint.

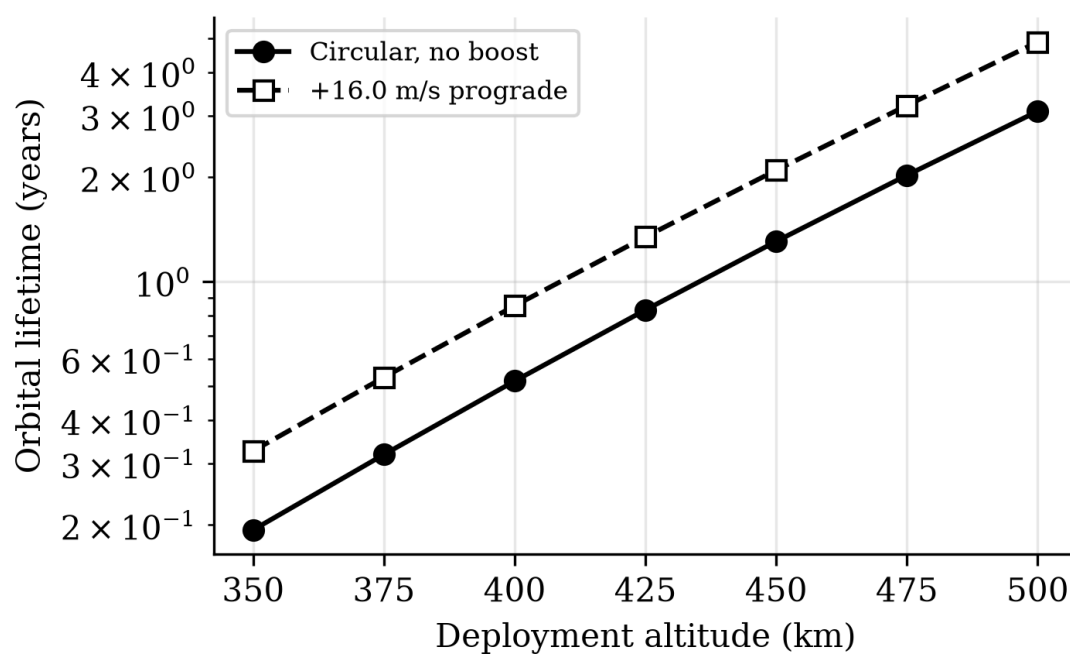


Figure 6: Modelled orbital-lifetime response. Absolute values depend on atmosphere.

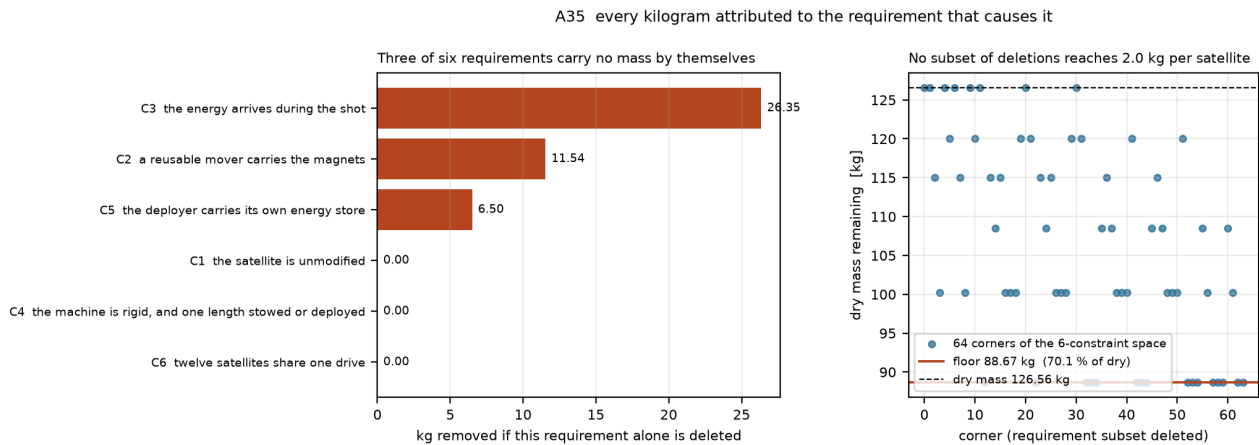


Figure 7: Constraint and mass-attribution analysis; mass is a model rollup.

Decision and evidence boundary

The completed analytical result has three parts. The model supports a commandable release and conditional orbital-energy benefit; the full-system 3U mass criterion fails; and the physical behavior remains unverified. The last category includes release-cycle reliability, shock and arrest loads transmitted to stowed satellites, field exposure near a customer payload, host attitude/control authority and provider approval. The known-problems register ranks several as potentially design-fatal. Their existence limits product claims but does not undo the completed analytical finding.

9. Conclusion

The Gen5 computational thesis reaches a complete engineering conclusion within its stated scope. Its system-level model predicts a commandable moderate-acceleration release and a conditional orbital-energy benefit. Complete installed-mass accounting then shows the configuration misses its preset 3U benchmark. The answer to the thesis trade question is therefore conditional benefit with an unfavourable 3U system cost. Hardware behavior and interfaces remain outside the tested evidence, so the result is a research conclusion rather than a flight-readiness claim.

The thesis's strongest engineering contribution is the transparent decision record: the full machine was modelled, independent numerical checks corrected several claims, and an unfavourable mass result was retained. That closes the Gen5 study and provides a defensible input to a separate future architecture trade.

10. Future work

Gen6's controlling objective is a reusable launch path that sequentially accepts different supported CubeSat sizes and commands each departure condition. The approximately 1-2 m/s to 1 km/s span is an investigation envelope; the upper end may force acceleration, energy, travel, payload or provider constraints that reject candidates. Electromagnetic systems are preferred for comparison but are not preselected. Competent springs, electromechanical/stored-energy, gas/fluid, hybrid, BOLLEY and conventional mission substitutes belong in the same comparison boundary.

The near-term milestone is one named test-article configuration that can be independently reviewed. Its package needs controlled geometry and drawings, a bill of materials, host and payload interfaces, full installed mass/energy/thermal accounting, assembly and inspection instructions, instrumented procedures, predetermined pass/fail and stop criteria, uncertainty and model-credibility evidence, retained failures, and an explicit build/test release decision. A rejected candidate is a valid research outcome; it must not be presented as a build-ready survivor.

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